Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *CRISISBENCH* | ***Context:*** *Argentina — Buenos Aires Policy Maker Cohort (Disaster Reporting Reliability)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is MODERATE priority per elicitation. The benchmark produces discrete classification labels evaluated via weighted precision/recall/F1 [Q75], with documented model families (CNN [Q63], fastText [Q64], BERT/DistilBERT/RoBERTa [Q66]) achieving 0.866 F1 informativeness and 0.829 F1 humanitarian on consolidated English test set [Q84]. Hyperparameters are fully documented [Q113–Q121] and ten re-runs are performed for stability [Q72]. The output form is clean, reproducible, and suitable for downstream pipeline construction. However, two structural gaps remain: (a) the deployment envisions an aggregated continuous credibility score, which the benchmark does not produce — it requires downstream construction (user-acknowledged); no published pipeline combining CrisisBench outputs with external metadata exists [WEB-21]; (b) multilingual evaluation shows ~8% F1 drop on humanitarian classification using BERT [Q99], indicating that even where multilingual output is attempted, performance degrades substantially. Pre-trained models are mainly trained on non-Twitter text, an acknowledged limitation [Q65]. The discrete-to-continuous transformation is a structural gap but is bounded and tractable.

ID	Evidence
Q113	“In this section, we report parameters for CNN and BERT model.” (p.11)
Q114	“All experimental scripts will be publicly Hyper-parameters include:” (p.11)
Q115	“Batch size: 8” (p.11)
Q116	“Number of epochs: 10” (p.11)
Q117	“Max seq length: 128” (p.11)
Q118	“Learning rate (Adam): 2e-5” (p.11)
Q119	“BERT (bert-base-uncased): L=12, H=768, A=12, total parameters = 110M; where L is number of layers (i.e., Transformer blocks), H is the hidden size, and A is the number of self-attention heads.” (p.11)
Q120	“DistilBERT (distilbert-base-uncased): it is a distilled version of the BERT model consists of 6-layer, 768-hidden, 12-heads, 66M parameters.” (p.11)
Q121	“RoBERTa (roberta-large): it is using the BERT-large architecture consists of 24-layer, 1024-hidden, 16-heads, 355M parameters.” (p.11)
Q75	“To measure the performance of each classifier, we use weighted average precision (P), recall (R), and F1-measure (F1). The rationale behind choosing the weighted metric is that it takes into account the class imbalance problem.” (p.7)
Q63	“The current state-of-the-art disaster classification model is based on the CNN architecture.” (p.6)
Q64	“For the fastText (Joulin et al. 2017), we used pre-trained embeddings trained on Common Crawl, which is released by fastText for English.” (p.6)
Q66	“In this work, we choose the pre-trained models BERT (Devlin et al. 2019), DistilBERT (Sanh et al. 2019), and RoBERTa (Liu et al. 2019) for the classification tasks.” (p.6)
Q84	“The model trained using the consolidated dataset achieves 0.866 (F1) for informativeness and 0.829 for humanitarian, which is better than the models trained using individual datasets.” (p.8)
Q72	“Due to the instability of the pre-trained models as reported in (Devlin et al. 2019), we perform ten re-runs for each experiment using different seeds, and we select the model that performs best on the dev set.” (p.6)
Q99	“We observe that performance dropped significantly for the humanitarian task compared to English-only dataset. For example, ~8% drop using BERT model.” (p.9)

Q65	“Though the pre-trained models are mainly trained on non-Twitter text, we hypothesize that their rich contextualized embeddings would be beneficial for the disaster domain.” (p.6)
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)

Strengths

- Standard, reproducible evaluation methodology with weighted F1 chosen explicitly to handle class imbalance [Q75]
- Multiple model families benchmarked (CNN, fastText, transformers) with full hyperparameter documentation [Q113–Q121] and stability via 10 re-runs [Q72]
- Discrete classification labels are easily consumed by downstream credibility-aggregation pipelines, supporting the user's planned architecture

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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Partial mismatch: benchmark produces discrete classification labels; deployment envisions an aggregated continuous credibility score combining model outputs with external metadata. The user acknowledges this requires downstream construction. No published end-to-end credibility pipeline applicable to the deployment exists [WEB-21].

OF-2: Not applicable — deployment is text-only; no text-to-speech requirement.

OF-3: Target cohort is high-literacy (master's degree+) [WEB-1, WEB-2]; literacy is not a constraint. Output form does not need accessibility adaptations for the analyst cohort.

OF-4: Documented form mismatches: (a) discrete labels vs. envisioned continuous credibility score (user-acknowledged downstream construction step); (b) ~8% F1 drop in multilingual humanitarian classification [Q99] — output reliability degrades for non-English content. These threaten external validity for the operational credibility-scoring use case but are bounded.

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Q84	"The model trained using the consolidated dataset achieves 0.866 (F1) for informativeness and 0.829 for humanitarian, which is better than the models trained using individual datasets." (p.8)
WEB-21	2024 NLP credibility survey: aggregating multiple credibility signals into a unified score remains fragmented and underspecified (https://arxiv.org/html/2410.21360v2)
WEB-1	Argentine national literacy 98.08% (INDEC); target cohort effectively 100% (https://www.perfil.com/noticias/sociedad/el-censo-argentino-arrojo-98-de-alfabetizacion-en-el-pais-pero-no-se-refleja-en-las-aulas.phtml)
WEB-2	https://es.theglobaleconomy.com/Argentina/Literacy_rate/
Q99	"We observe that performance dropped significantly for the humanitarian task compared to English-only dataset. For example, ~8% drop using BERT model." (p.9)
Q65	"Though the pre-trained models are mainly trained on non-Twitter text, we hypothesize that their rich contextualized embeddings would be beneficial for the disaster domain." (p.6)

Information Gaps

- Whether downstream aggregation of CrisisBench classifier outputs with external Argentine source-metadata produces operationally useful credibility scores
- Whether the ~8% multilingual F1 drop generalizes to or worsens for Rioplatense Spanish

Requires Expert Verification

- Validation by Argentine policy makers of what continuous credibility score thresholds are operationally meaningful for their decision-making

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Remediation

Gap 1: Discrete labels do not natively yield the aggregated continuous credibility score the deployment envisions; ~8% F1 drop in multilingual setting

Recommendation: Design the downstream credibility-aggregation function explicitly: define how classifier confidence scores combine with source-popularity, institutional-affiliation, and corroboration metadata to yield a continuous score. Validate score thresholds with AMBA policy makers and document the transformation pipeline for transparency under Argentine AI governance instruments [WEB-11, WEB-12].

ID	Evidence
WEB-11	https://regulations.ai/regulations/argentina-summary
WEB-12	https://digital.nemko.com/regulations/ai-regulation-argentina